

AMERICAN LATIN CLASS ATTENDANCE SYSTEM

Requirements

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Requirements

Functional Requirements

ID	Requirement	Priority	Acceptance Criteria
FR-01	Public landing page shows academy information, branches, styles, levels, contact/enrollment request.	High	Visitor can open <code>/</code> , read branches/styles/levels and submit an enrollment request.
FR-02	Google OAuth registration/login with internal roles.	High	Backend verifies Google ID token in production, links <code>google_sub</code> , creates app session and never trusts Google for app roles.
FR-03	Secure session management.	High	Session cookie is HttpOnly/SameSite, logout revokes server session, expired sessions return 401, private pages redirect after logout/back button.

ID	Requirement	Priority	Acceptance Criteria
FR-04	Role-based authorization.	High	Student/Teacher/BranchDirector/GeneralDirector/ access is enforced by middleware and tested.
FR-05	Branch, student, teacher, class group and class session management.	High	Directors/Admin can create/update core academic authenticated users can list/read.
FR-06	Dance category/style management.	Medium	Urban, Tropical and Ethnic styles are seeded and manageable through API.
FR-07	Student attendance.	High	Teacher/director records one attendance status per student and class session. Duplicates are rejected.
FR-08	Teacher check-in/check-out.	High	Teacher/director opens one active check-in and later closes it for hour calculation.
FR-09	Absence justifications.	Medium	Authorized users submit a justification and director approve/reject it.
FR-10	Scholarship eligibility and evaluation.	High	System marks candidates at $\geq 90\%$ attendance over two months; directors register theory/practice results and approval decision.
FR-11	Level promotion eligibility and evaluation.	High	B1 students can be marked as B2 candidates based on attendance and directors register consistency/theory/practice results.
FR-12	Teacher worked hours and payment report.	Medium	Completed check-in/out records produce hours, hourly rate and amount.
FR-13	Branch and consolidated reports.	Medium	GeneralDirector/Admin can view branch summaries and restricted reports.
FR-14	Audit logs.	Medium	Administrative and attendance actions are recorded and visible to GeneralDirector/Admin.

Non-Functional Requirements

ID	Requirement	Evidence
NFR-01	REST API versioned with <code>/api/v1</code> .	<code>06Code/src/routes/api.js</code>

ID	Requirement	Evidence
NFR-02	Clean Code, MVC, services and repositories.	Controllers delegate to services; persistence is isolated by repository classes.
NFR-03	PostgreSQL normalized schema and seeders.	<code>06Code/migrations/001_initial_schema.sql</code> , <code>06Code/seeders/001_seed.sql</code>
NFR-04	No secrets in repository.	<code>.env.example</code> only contains placeholders.
NFR-05	Restricted CORS.	<code>CORS_ORIGINS</code> env configuration.
NFR-06	Auth rate limiting.	<code>express-rate-limit</code> on <code>/auth/google</code> .
NFR-07	Consistent JSON responses.	<code>ApiResponse</code> success/error envelope.
NFR-08	Centralized validation.	Zod validators under <code>src/validators</code> .
NFR-09	Consistent error handling.	<code>AppError</code> and <code>errorHandler</code> middleware.
NFR-10	AWS deployment preparation.	EC2/RDS/ALB guide and PlantUML deployment diagram.

User Stories

ID	Story
US-01	As a Visitor, I want to request enrollment from the public page so the academy can contact me.
US-02	As a Student, I want secure access to my private academic system session.
US-03	As a Teacher, I want to record student attendance and check in/out for my worked hours.
US-04	As a BranchDirector, I want to manage students, teachers, classes and absence justifications for my branch.
US-05	As a BranchDirector, I want to review scholarship and promotion candidates before registering evaluations.
US-06	As a GeneralDirector, I want consolidated branch reports and audit logs.

ID	Story
US-07	As an Admin, I want to assign internal roles and seed permissions without depending on Google profile data.

Role/Permission Matrix

Feature	Visitor	Student	Teacher	BranchDirector	GeneralDir
View landing page	X	X	X	X	
Submit enrollment request	X				
Login/logout/session		X	X	X	
List own/private data		X	X	X	
Record student attendance			X	X	
Teacher check-in/out			X	X	
Submit absence justification		X	X	X	
Review absence justification				X	
Manage academic data				X	
Scholarship/promotion evaluation				X	
Branch reports				X	
Consolidated reports					
Audit logs					
Assign roles					

Business Rules

- BR-01: Scholarship candidate requires at least 90% attendance in a two-month period.
- BR-02: Scholarship approval is never automatic; directors must register theory and practice evaluation results.
- BR-03: Scholarship percentages allowed are 25%, 50%, 75% and 100%.

- BR-04: Level promotion applies to B1 students moving to B2.
- BR-05: Level promotion approval requires attendance evidence, consistency score, theory score and practice score.
- BR-06: A student can have only one attendance record per class session.
- BR-07: A teacher can have only one open check-in at a time.
- BR-08: Roles are owned by the application and are not inferred from Google claims.
- BR-09: Private pages and APIs must use `Cache-Control: no-store`.
- BR-10: Administrative and academic state-changing actions must be audited.

Prioritized Backlog

1. Auth/session, RBAC, no-store private pages and tests.
2. PostgreSQL schema, seeders and repository abstraction.
3. Branch/student/teacher/style/class management APIs.
4. Student attendance and teacher check-in/check-out APIs.
5. Absence justifications and audit logs.
6. Scholarship and level promotion candidate/evaluation workflows.
7. Reports by branch, teachers and consolidated roles.
8. Landing page, enrollment form and private dashboard shell.
9. AWS EC2/RDS/ALB deployment documentation and diagrams.
10. Final test evidence and requirement reevaluation.

Requirement-To-Endpoint Traceability

Requirement	Endpoints
FR-01	<code>GET /</code> , <code>POST /api/v1/enrollment-requests</code>
FR-02	<code>POST /api/v1/auth/google</code> , <code>GET /api/v1/users</code> , <code>PATCH /api/v1/users/{id}/role</code>
FR-03	<code>GET /api/v1/auth/me</code> , <code>POST /api/v1/auth/logout</code> , <code>/private/dashboard.html</code>
FR-04	All protected endpoints through <code>requireAuth</code> / <code>allowRoles</code>
FR-05	<code>/api/v1/branches</code> , <code>/api/v1/students</code> , <code>/api/v1/teachers</code> , <code>/api/v1/class-groups</code> , <code>/api/v1/class-sessions</code>
FR-06	<code>/api/v1/dance-categories</code> , <code>/api/v1/dance-styles</code>
FR-07	<code>POST /api/v1/student-attendance</code>
FR-08	<code>POST /api/v1/teacher-attendance/check-in</code> , <code>PATCH /api/v1/teacher-attendance/{id}/check-out</code>

Requirement	Endpoints
FR-09	<code>/api/v1/absence-justifications</code> , <code>/api/v1/absence-justifications/{id}/review</code>
FR-10	<code>GET /api/v1/reports/scholarships/{studentId}/candidate</code> , <code>/api/v1/scholarship-evaluations</code>
FR-11	<code>GET /api/v1/reports/level-promotions/{studentId}/candidate</code> , <code>/api/v1/level-promotion-evaluations</code>
FR-12	<code>GET /api/v1/reports/teachers/{teacherId}/payment</code>
FR-13	<code>GET /api/v1/reports/branches/summary</code>
FR-14	<code>GET /api/v1/audit-logs</code>